A Counterexample to Polynomial-Fiber Covering

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Status

This packet gives a likely valid counterexample to the broad polynomial-fiber covering question raised in Ortega Castillo–Prieto [1]. It does not settle the polynomial cluster value theorem itself. It shows that, for the standard choice $X = \ell_2$ and $H(B) = H^\infty(B_X)$, the polynomial fibers over points of $\overline{B}_{X^{**}}$ need not cover the maximal ideal space of $H(B)$.

Source Problem Evidence

The source passage is in Section 3 of [1]. In the notation of the source paper, it asks whether

$$\bigcup_{x_0^{**} \in \overline{B}^{**}} M^{\mathcal{P}}_{x_0^{**}}(H(B)) = M_{H(B)}.$$

polynomial cluster value problem, we have found some features of the latter that up to now make it distinctive. In this section we focus on the polynomial cluster value problem for the algebra $H^\infty(B)$ for B the ball of a space of continuous functions.

For the original cluster value problem, the size and structure of fibers for some spectra of subalgebras of $H^\infty(B)$ have been recently studied in [**3**] and [**6**]. In contrast with the case of the original cluster value problem, we do not know if the union of the polynomial fibers $M^{\mathcal{P}}_{x_0^{**}}$, with $x_0^{**} \in \bar{B}^{**}$, is all of $M_{H(B)}$. Still, the polynomial cluster value problem is in general nontrivial, as illustrated by the size of the polynomial fibers in the following example [**19**, pp. 1565-1567].

Example 3.1 (Johnson, Ortega Castillo). If K is an infinite compact Hausdorff space then each of the polynomial fibers $M^{\mathcal{P}}_{f_0}$, for $f_0 \in B_{C(K)}$ and the algebra $H^\infty(B_{C(K)})$ contains a holomorphic copy of B_{ℓ}

Definitions and Scope

Let X be a complex Banach space and $B = B_X$. The algebra $A_u(B)$ is the uniform algebra of bounded holomorphic functions on B that are uniformly continuous on B. For a uniform algebra $H(B)$ with

$$A_u(B) \subset H(B) \subset H^\infty(B),$$

the source paper defines the polynomial fiber over $x^{**} \in \overline{B}^{**}$ by

$$M^{\mathcal{P}}_{x^{**}}(H(B)) = \{\tau \in M_{H(B)} : \tau|_{A_u(B)} = \delta_{x^{**}}\},$$

where $\delta_{x^{**}}$ denotes evaluation at x^{**} through Aron–Berner extension on $A_u(B)$.

Thus an element of $M_{H(B)}$ lies in the union of the polynomial fibers only if its restriction to $A_u(B)$ is one of these evaluation characters.

Proof Intuition

The obstruction is already visible on the Hilbert ball. Take evaluations at the points re_n in B_{ℓ_2}, with $0 < r < 1$. Any cluster point of these evaluations on $H^\infty(B_{\ell_2})$ is a character. On all linear functionals it looks like evaluation at 0, because e_n is weakly null. But the continuous quadratic polynomial

$$P(x) = \sum_{n=1}^{\infty} x_n^2$$

has value r^2 at every re_n, while $P(0) = 0$. Therefore the restriction of the cluster character to $A_u(B_{\ell_2})$ cannot be any evaluation character. Consequently the cluster character cannot belong to any polynomial fiber.

Theorem 1. *Let $X = \ell_2$, let $B = B_X$, and let $H(B) = H^\infty(B)$. Then*

$$\bigcup_{x \in \overline{B}_{\ell_2}} M_x^{\mathcal{P}}(H^\infty(B)) \subsetneq M_{H^\infty(B)}.$$

In particular, the polynomial fibers over points of $\overline{B}_{X^{**}}$ do not necessarily cover the maximal ideal space of $H(B)$.

Proof. Fix $0 < r < 1$, and let e_n be the standard unit vector basis of ℓ_2. For every n, point evaluation δ_{re_n} is an element of $M_{H^\infty(B)}$. Since $M_{H^\infty(B)}$ is compact in the weak-star topology, the net (or, equivalently, a subnet of the sequence) (δ_{re_n}) has a cluster point; call it $\tau \in M_{H^\infty(B)}$.

Let $\varphi = \tau|_{A_u(B)}$. We claim that φ is not an evaluation character on $A_u(B)$. First, for every continuous linear functional $x^* \in \ell_2^*$, writing $x^*(e_n) = a_n$ with $(a_n) \in \ell_2$, we have $a_n \to 0$. Hence

$$\varphi(x^*) = \lim \delta_{re_n}(x^*) = \lim r x^*(e_n) = 0.$$

Therefore, if $\varphi = \delta_x$ for some $x \in \overline{B}_{\ell_2}$, then every linear functional vanishes at x, so $x = 0$.

Now define

$$P(x) = \sum_{j=1}^{\infty} x_j^2, \qquad x = (x_j) \in \ell_2.$$

This series is absolutely convergent and

$$|P(x)| \le \sum_j |x_j|^2 = \|x\|_2^2.$$

Moreover,

$$|P(x) - P(y)| = \left| \sum_j (x_j - y_j)(x_j + y_j) \right| \le \|x - y\|_2 \, \|x + y\|_2,$$

so P is a continuous 2-homogeneous polynomial and belongs to $A_u(B)$. Along the chosen evaluations,

$$P(re_n) = r^2 \quad \text{for all } n.$$

Thus

$$\varphi(P) = \tau(P) = r^2,$$

while $\delta_0(P) = P(0) = 0$. This contradicts the only possible evaluation case identified above. Hence φ is not equal to δ_x for any $x \in \overline{B}_{\ell_2}$.

Because ℓ_2 is reflexive, $\overline{B}_{X^{**}}$ is identified with $\overline{B}_{\ell_2}$. If τ belonged to a polynomial fiber $M_x^{\mathcal{P}}(H^\infty(B))$, then by definition its restriction to $A_u(B)$ would be δ_x. But $\tau|_{A_u(B)} = \varphi$ is not any evaluation character. Therefore

$$\tau \notin \bigcup_{x \in \overline{B}_{\ell_2}} M_x^{\mathcal{P}}(H^\infty(B)).$$

This proves the strict inclusion. □

Verification Notes

The proof uses only:

- compactness of the maximal ideal space of a unital commutative Banach algebra;
- weak nullity of the standard basis of ℓ_2;
- the explicit continuous polynomial $P(x) = \sum_j x_j^2$;
- the definition of polynomial fiber as a restriction fiber over evaluation on $A_u(B)$.

No computational verification is needed.

Limitations

This is not a counterexample to the polynomial cluster value theorem

$$Cl_B^{\mathcal{P}}(f, x^{**}) = \widehat{f}(M_{x^{**}}^{\mathcal{P}})$$

for a fixed fiber. It only disproves the broader structural hope that the polynomial fibers cover all of $M_{H(B)}$. It also does not resolve the more specialized $C(K)$ questions pursued in Section 3 of [1].

Novelty and Literature Check

Before promotion, I searched the local run indexes and parsed arXiv sources for the arXiv id 1702.06471, the title phrase "polynomial cluster value problem", and the phrases "union of the polynomial fibers", "polynomial fibers", and "range of $\pi^{\mathcal{P}}$". I also searched the web for exact phrases around the covering question. I found the source paper and later related literature on large fibers of $M(A_u(B_{\ell_p}))$, notably Dimant–Lassalle–Maestre [2], but I did not find an exact statement of the $H^\infty(B_{\ell_2})$ polynomial-fiber covering counterexample above.

The construction is closely related in spirit to the known phenomenon that maximal ideal spaces of $A_u(B_X)$ can contain many non-evaluation characters. For that reason, the novelty confidence should be treated as moderate: the packet gives a direct, self-contained counterexample to the specific covering question, but it uses standard compactness and basis-vector ideas rather than a new technical method.

Human Review Recommendation

Review the definition of $M_x^{\mathcal{P}}(H(B))$ against the source paper and check that the restriction of the constructed τ to $A_u(B)$ is indeed not an evaluation character. The key audit line is the simultaneous use of all linear functionals to force a hypothetical evaluation point to be 0, followed by the quadratic polynomial P separating the character from δ_0.

References

[1] S. Ortega Castillo and A. Prieto, *The polynomial cluster value problem*, arXiv:1702.06471, 2017. https://arxiv.org/abs/1702.06471

[2] V. Dimant, S. Lassalle, and M. Maestre, *Fibers and Gleason parts for the maximal ideal space of $\mathcal{A}_u(B_{\ell_p})$*, arXiv:2409.13889, 2024. https://arxiv.org/abs/2409.13889